



Smart Restaurant

Restaurant Management System

From the order to the kitchen, from stock to reports — the whole restaurant in one application.

Single-file setup

Works offline

Turkish / English

The problem it solves

Most restaurants run on three separate things: a till program, paper in the kitchen, and a spreadsheet for stock.

- A gap between the check and the kitchen — wrong and late orders
- Finding out an ingredient has run out only when the order is placed
- Real profit stays invisible because recipe cost is unknown
- Reports collected at month end, too late to act on
- No trace of who voided or discounted what

The system at a glance

One application, one single-file install.

14

modules

from POS to backups

12

user roles

62 separate permissions

484

automated tests

run on every release

2

languages

Turkish and English

Modules

Each module stands on its own; together they produce the real value.

POS and orders

Touch sales, split checks, multiple payments, coupons and discounts

Kitchen display

Live ticket flow, split by station, time alerts

Stock and recipes

Batch tracking, FIFO/FEFO, automatic deduction, waste records

Floor and reservations

Table plan, merging, waiting list

Customers and loyalty

Segments, points, campaigns, consent records

Staff

Shifts, timesheets, leave, performance

Reporting

Sales, profitability, daily closing, Excel/PDF output

Statistics centre

Period comparison, intensity matrix

Artificial intelligence

Local or cloud model, smart analyses

Backups

Consistent snapshot, safe restore

Training

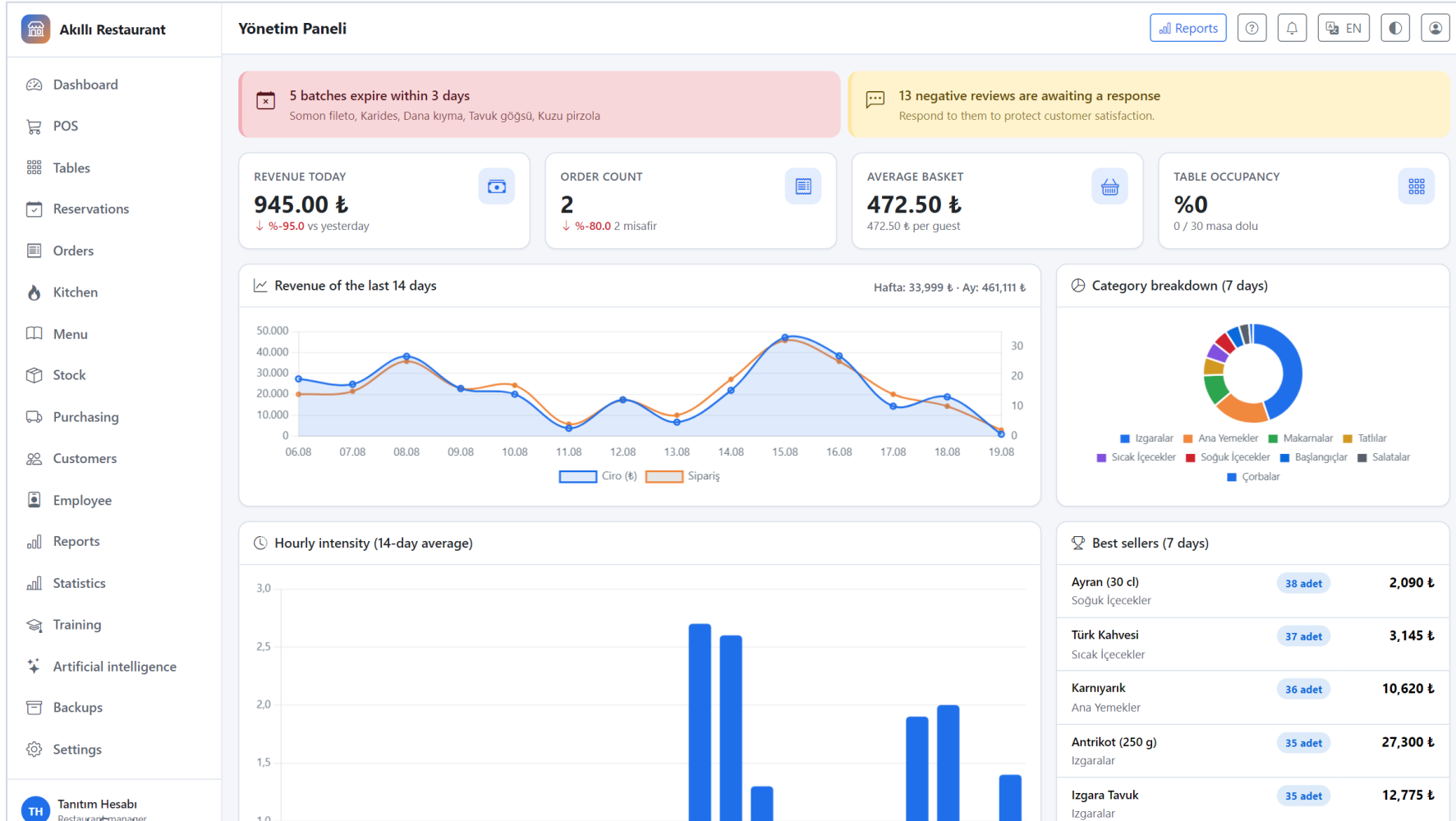
In-app guide, lessons by role

Security

Role-based permissions, audit log, PIN approval

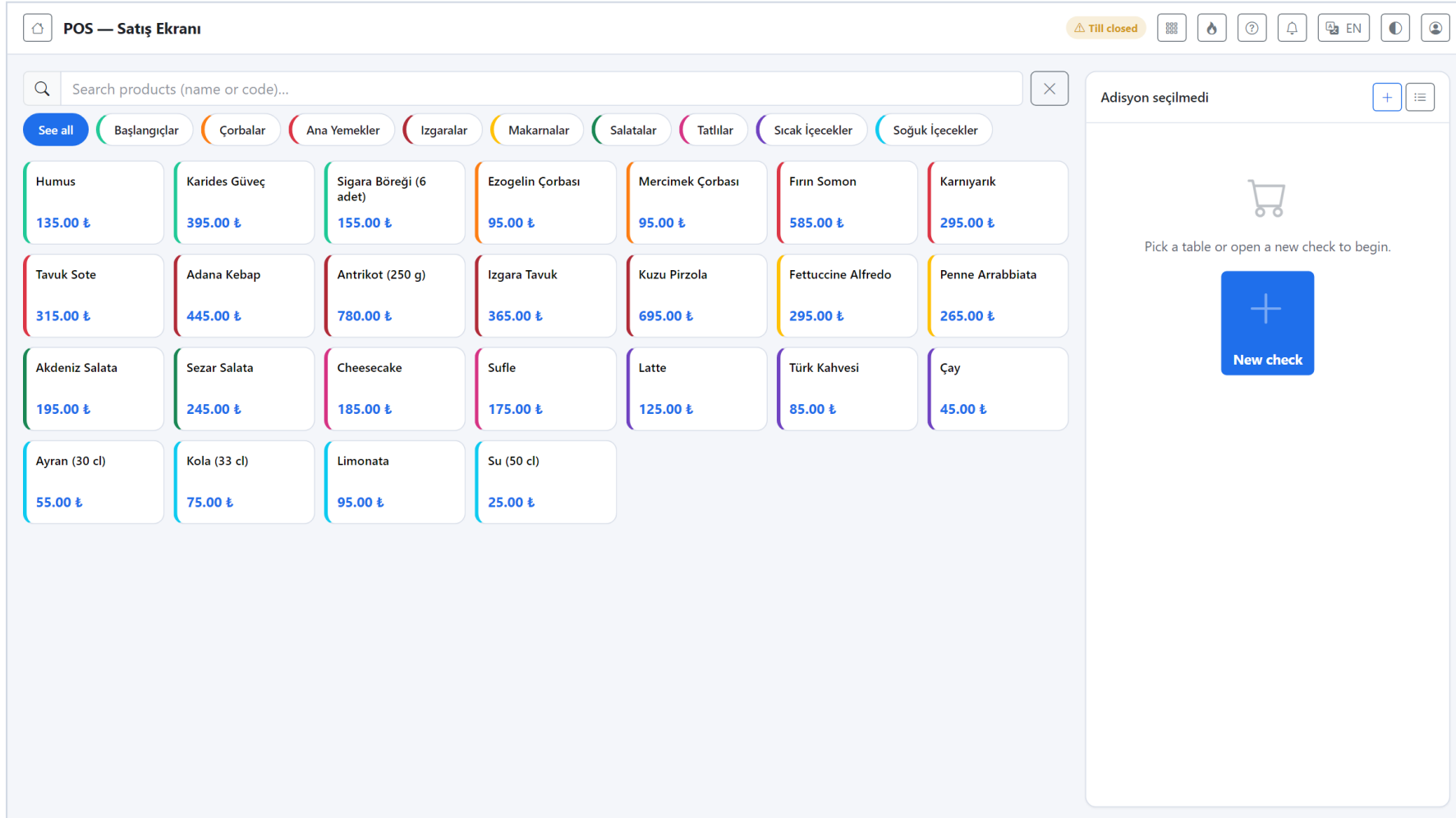
Dashboard

Today's revenue, order count, occupancy, 14-day trend, category breakdown and critical alerts on one screen.



POS — sales screen

Touch-friendly selling: category filters, quick product search, check and table management.



Kitchen display (KDS)

Orders split by station and stream live; overdue tickets change colour and sound an alert.

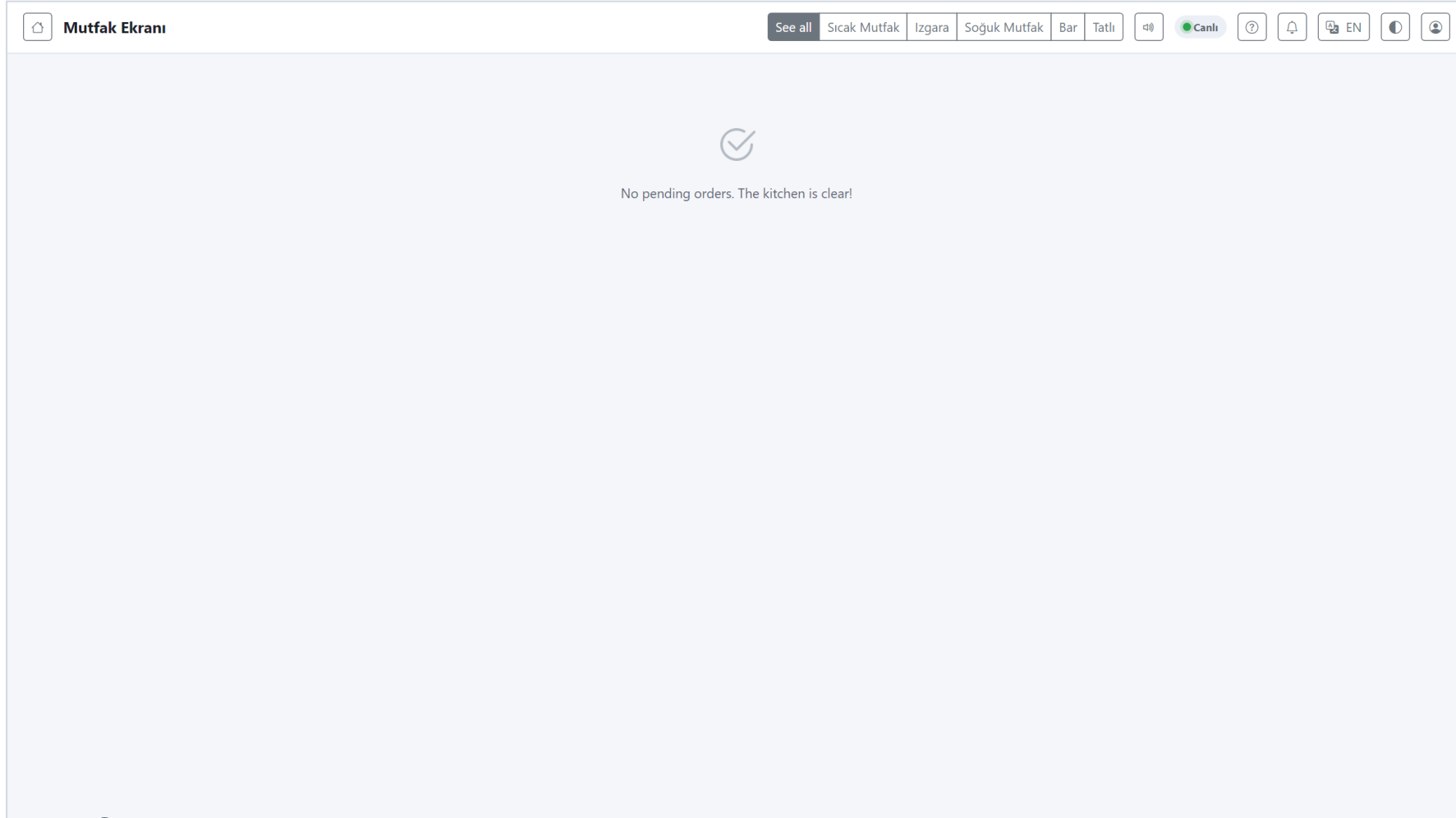
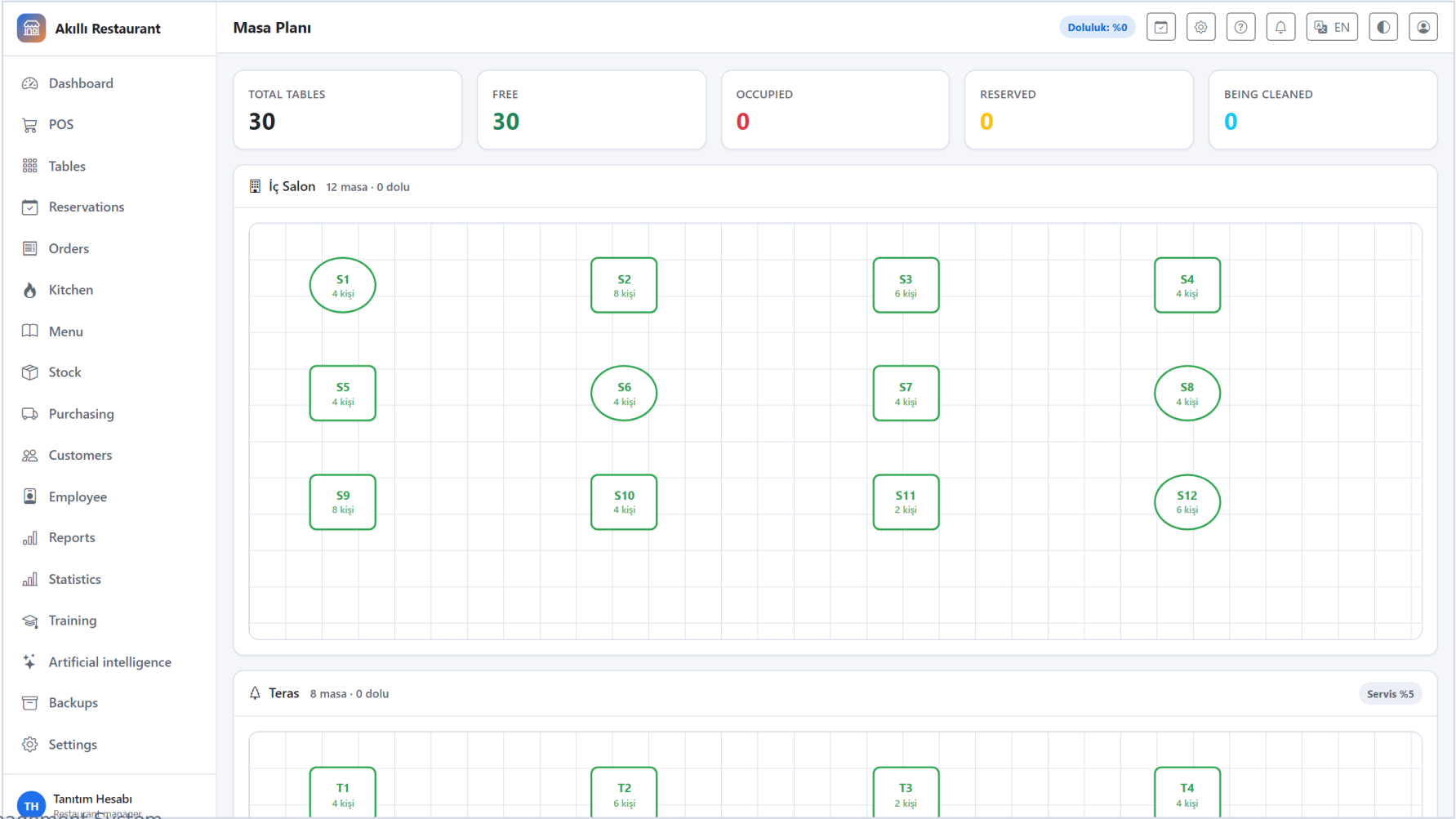


Table plan

Floor sections, table states and QR-coded table management at a glance.



Stock and ingredients

Lot-based FIFO/FEFO tracking, critical-level alerts and run-out forecasts.

Akıllı Restaurant

Dashboard

POS

Tables

Reservations

Orders

Kitchen

Menu

Stock

Purchasing

Customers

Employee

Reports

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Backups

Settings

TH

Tanıtım Hesabı

Restaurant management

Stok Yönetimi

Warnings

+ New ingredient

?

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INGREDIENT VARIETY

32

STOCK VALUE

96,079 ₺

CRITICAL LEVEL

0

EXPIRING SOON

10

within 7 days

Category

See all

Stock status

See all

Search

Ingredient name or code

Filter

INGREDIENT	CURRENT	CRITICAL	TIER	AVG. COST	VALUE	ESTIMATED DAYS	ACTION
Ayran (30cl) İçecek · FEFO - first expired, first out	133.422 adet	36	Normal	8.50	1,134.09	913 gün	
Beyaz peynir Süt Ürünleri · FEFO - first expired, first out	14,101.759 g	3,000	Normal	0.24	3,384.42	—	
Biber Sebze ve Meyve · FEFO - first expired, first out	9,917.406 g	2,000	Normal	0.05	495.87	—	
Dana antrikot Et ve Tavuk · FEFO - first expired, first out	22,636.028 g	4,000	Normal	0.68	15,392.50	—	
Dana kıyma Et ve Tavuk · FEFO - first expired, first out	24,965.285 g	5,000	Normal	0.42	10,485.42	3,816 gün	
Domates Sebze ve Meyve · FEFO - first expired, first out	21,318.747 g	4,000	Normal	0.04	852.75	—	
Kahve çekirdeği İçecek · FIFO - first in, first out	8,238.169 g	1,500	Normal	0.72	5,931.48	767 gün	
Karabiber Baharat · FIFO - first in, first out	1,230.511 g	300	Normal	0.85	1,045.93	1,735 gün	
Karides Deniz Ürünleri · FEFO - first expired, first out	8,280.547 g	1,500	Normal	0.95	7,866.52	1,128 gün	
Kaşar peyniri	15,667.024 g	3,000	Normal	0.32	5,013.45	751 gün	

Menu and recipes

Product cards, portions and modifiers; real cost and margin derived from the recipe.

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Tanıtım Hesabı

Restaurant manager

Menü Yönetimi

Categories

+ New product

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TOTAL PRODUCTS

25

ACTIVE

25

NOT FOR SALE

0

NO RECIPE

0

Cost cannot be calculated

Category

See all

Status

See all

Search

Product name or code

Filter

PRODUCT	CATEGORY	PRICE	COST	MARGIN	STATION	ALLERGEN	STATUS	ACTION
Humus URN1438 · 6 dk	Başlangıçlar	135.00 ₺	21.70	%82	Soguk Mutfak	1	On sale	👁 ☰ ✎ ⏸
Karides Güveç URN3545 · 14 dk	Başlangıçlar	395.00 ₺	181.62	%49	Sıcak Mutfak	2	On sale	👁 ☰ ✎ ⏸
Sigara Böreği (6 adet) URN3941 · 10 dk	Başlangıçlar	155.00 ₺	49.08	%65	Sıcak Mutfak	2	On sale	👁 ☰ ✎ ⏸
Ezogelin Çorbası URN9546 · 5 dk	Çorbalar	95.00 ₺	27.47	%68	Sıcak Mutfak	1	On sale	👁 ☰ ✎ ⏸
Mercimek Çorbası URN3777 · 5 dk	Çorbalar	95.00 ₺	25.56	%70	Sıcak Mutfak	1	On sale	👁 ☰ ✎ ⏸
Fırın Somon URN6514 · 16 dk	Ana Yemekler	585.00 ₺	193.48	%64	Sıcak Mutfak	1	On sale	👁 ☰ ✎ ⏸
Karniyank URN8336 · 22 dk	Ana Yemekler	295.00 ₺	82.51	%69	Sıcak Mutfak	—	On sale	👁 ☰ ✎ ⏸
Tavuk Sote URN4462 · 14 dk	Ana Yemekler	315.00 ₺	63.59	%78	Sıcak Mutfak	—	On sale	👁 ☰ ✎ ⏸
Adana Kebap URN1638 · 18 dk	Izgalar	445.00 ₺	93.29	%77	Izgara	—	On sale	👁 ☰ ✎ ⏸

Smart Restaurant Management System

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Customers and loyalty

Segments, loyalty points, consent records and permission-based masking.

Akıllı Restaurant

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Tanıtım Hesabı

Restaurant manager

Müşteriler

Reviews

New customer

?

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TOTAL CUSTOMERS

60

VIP

10

CHURN RISK

0

TOTAL REVENUE

376,987 ₺

Segment

See all

Search

Name, phone, customer code or company

Filter

CUSTOMER	CONTACT	SEGMENT	TIER	VISITS	TOTAL SPENDING	POINTS	LAST VISIT	RISK
Kerem Koç MZPVFFSU - Koç Ticaret Ltd. Allergy	0000****48	Corporate	Silver	5	6,740.75	690	19.08.2026	Low
Selin Balcı MQSES6Y4 Allergy	0000****50	Regular	Silver	4	6,582.00	656	19.08.2026	Low
Onur Kaya MC3VS63M Allergy	0000****44	Regular	Silver	4	5,643.75	580	19.08.2026	Low
Elif Koç MSU5PSXZ Allergy	0000****58	Regular	Silver	3	5,756.25	574	19.08.2026	Low
Nur Erdem MBZST2T4	0000****20	Regular	Silver	3	4,804.81	497	19.08.2026	Low
Emre Erdem MNKD7679 Allergy	0000****08	VIP	Gold	7	10,393.00	1,090	19.08.2026	Low
İrem Kaya MY6SMUTG Allergy	0000****23	Regular	Silver	4	5,939.50	592	19.08.2026	Low
Mehmet Öztürk	0000****34	Regular	Silver	4	3,060.00	305	19.08.2026	Low

Reservations

Reservation list, waiting list and conflict checks.

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Tanıtım Hesabı

Restaurant Manager

Rezervasyonlar

+ New reservation

?

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RESERVATIONS TODAY

2

EXPECTED GUESTS

8

AWAITING APPROVAL

0

NO-SHOW (30 DAYS)

2

Date

dd.mm.yyyy

📅

Status

See all

Search

Name, code or phone

Filter

CODE	DATE / TIME	GUEST	GUEST	TABLE	STATUS	ACTION
R8QKNT5	19.08.2026 12:30 90 dk	Burak Balcı 0000***28	4	S12	Approved	👤 👤 ✎
RDPMSQU	19.08.2026 19:00 90 dk	Ece Öztürk 0000***02 Doğum günü	4	S10, T4	Approved	👤 👤 ✎
RN38856	20.08.2026 12:30 90 dk	Zeynep Demir 0000***43 İş yemeği	4	S1, S7	Approved	👤 👤 ✎
R9SVGNE	20.08.2026 13:00 90 dk	Ali Çelik 0000***01	2	S1	Approved	👤 👤 ✎
RG3WD55	20.08.2026 21:00 90 dk	Burak Balcı 0000***28	2	T1	Approved	👤 👤 ✎
RGTRR2J	21.08.2026 12:00 90 dk	Onur Çelik 0000***13	6	S6	Approved	👤 👤 ✎
RT5Q4Q7	21.08.2026 19:30 90 dk	Elif Koç 0000***30	3	V4	Approved	👤 👤 ✎
R4KWPM5	21.08.2026 20:00 90 dk	Emre Doğan 0000***09 Yıldönümü	6	S8	Approved	👤 👤 ✎
RRGF6NX	21.08.2026 21:00 90 dk	Kerem Şahin 0000***21	4	B6	Approved	👤 👤 ✎

Waiting list

Guest name

2

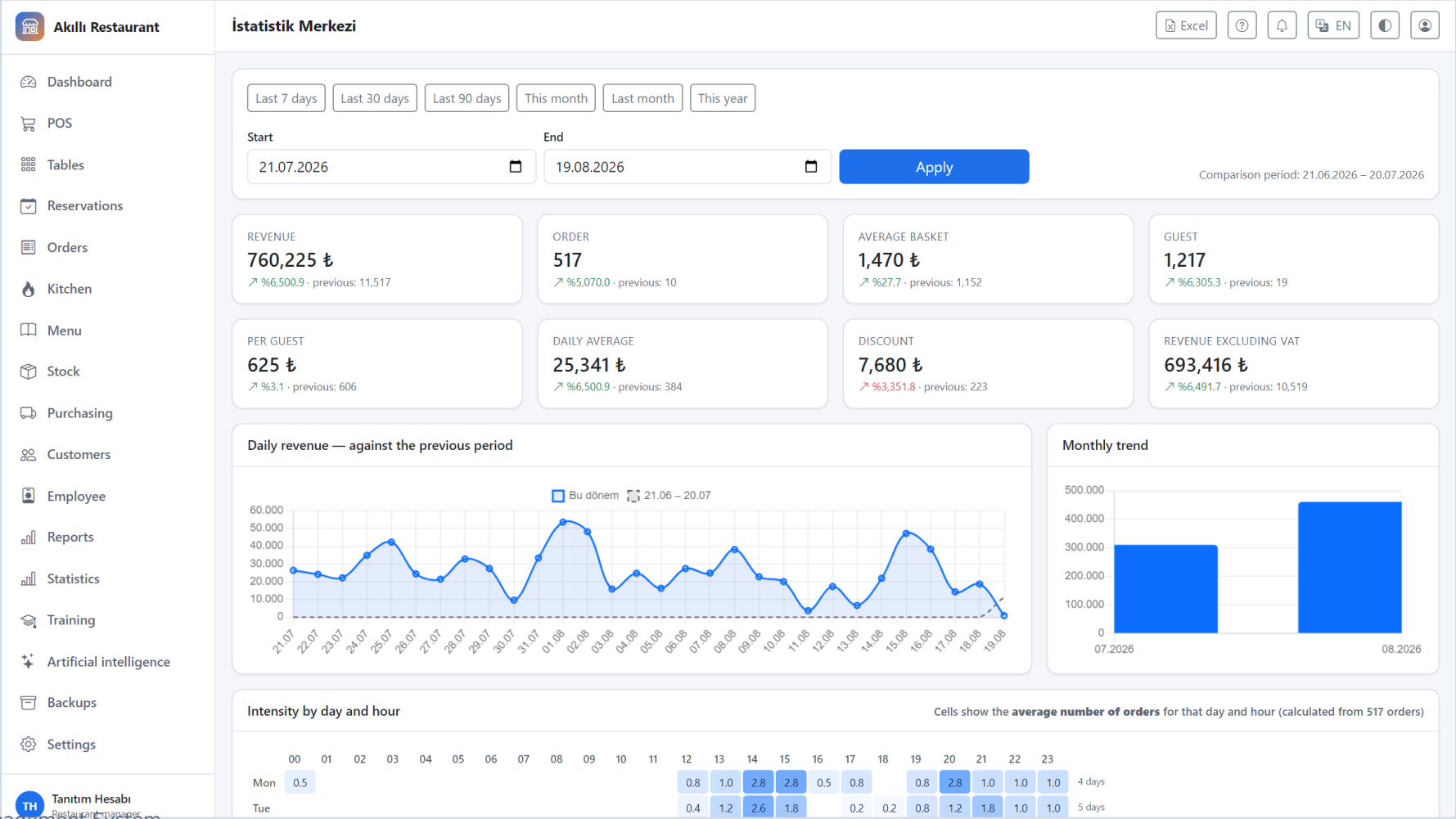
Phone

Add

No guests waiting.

Statistics centre

Period comparison, intensity matrix and trend charts.



AI assistant

Q&A over your own data; with a local model, nothing leaves the machine.

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Tanıtım Hesabı

Restaurant manager

Yapay Zekâ Asistanı

Model

Provider

Automatic (local first)

Task type

Reasoning and reporting

Providers marked are local; data never leaves your computer.

Budget

Today

0.0000 / 1.00 USD

This month

0.0000 / 20.00 USD

Local models are free and are not charged to the budget.

Conversations

+ New conversation

Restaurant assistant

ready

Restoranınızla ilgili bir soru sorun.
Your system data is added to the answer automatically.

Bugün en çok hangi ürün satıldı?

Bu hafta cirom geçen haftaya göre nasıl?

Hangi malzemeler kritik seviyede?

En kârlı 5 ürünüm hangileri?

Bugün kaç sipariş iptal edildi ve neden?

Yarın için kaç kişilik personel planlamalıyım?

Type your question... (e.g. which product brought the most profit this week?)

ⓘ

AI answers are based on your system data but can be wrong. Verify the relevant report before making an important decision. Customer personal data is masked automatically before the prompt is sent.

The order is entered once

- The server picks the product on the POS and adds a note
- "Send to kitchen" splits the order across stations: hot kitchen, grill, bar, pastry
- At the same moment the recipe's ingredients leave stock
- The kitchen screen shows the ticket instantly — no page refresh
- When the time is exceeded the ticket turns yellow, then red, with an audible alert

WHY IT MATTERS

Paper tickets, shouting across the pass and the "did it get through?" uncertainty disappear. Stock stops being a separate job; the sale itself updates it.

Seeing the real profit

- Every product has a recipe and a portion yield
- Ingredient cost is calculated per batch, at the real purchase price
- The profitability report gives profit, margin and food cost ratio per product
- Products without a recipe are listed separately as a warning
- The effect of a price change on margin can be simulated beforehand

AN HONEST NUMBER

A product without a recipe counts as zero cost, so its profit looks higher than it is. The system does not hide this; it says at the top of the report how many products lack a recipe.

Statistics centre

- The chosen period is compared side by side with the previous equal period
- A day × hour intensity matrix — for staff planning
- Customer behaviour: new vs returning, most valuable customers
- Waste rate, stock value, service speed and table turnover
- Every section exports to a single Excel file

NO DECISIONS ON THIN DATA

Cells resting on fewer than three observations are faded. A single busy Saturday is not presented as a regular pattern — the system tells you how much data each number rests on.

AI — without your data leaving the building

- Choose a local model (LM Studio, Ollama) or a cloud provider
- Sensitive data can be restricted to the local model only
- Customer personal data is masked before the prompt is sent
- Demand forecasting, menu engineering, waste and anomaly analysis
- A daily management summary: what went well, what needs attention

A FORECAST IS NOT A CERTAINTY

Every AI output is shown with its confidence level, the amount of data behind it and its limitations. A budget cap can be set; local models are free and are not charged against it.

Data safety and backups

- One-click backup from inside the app: database, files and a portable dump
- A consistent snapshot even while the system is running
- Integrity verification with SHA-256
- An automatic safety backup before any restore; a confirmation phrase is required
- Optional scheduled automatic backups

API KEYS STAY OUT OF THE BACKUP

Backup archives usually travel by email or cloud storage. Keys are excluded by default so they cannot leak that way; they can be included explicitly, and the archive is then marked.

Who sees what

Permissions are role-based; the menu shows only the sections you can actually use.

Role	What they see	Requires approval
Server	POS, tables, their own orders	Voids, discounts
Cashier	POS, payment, till, basic reports	Refunds, price overrides
Chef	Kitchen, menu, recipes, stock	—
Storekeeper	Stock, counts, purchasing, suppliers	—
Manager	Everything + financial reports + backups	—
Owner	Everything + restore + users	—

Security

Restaurant data means customer personal data. The system treats it that way.

- Role-based permissions, with per-user grants and revocations
- Manager PIN approval for voids, refunds and discounts
- An immutable audit log: who, when, what
- Account lockout after five failed sign-ins
- Customer phone and email are masked for users without the permission
- Data erasure and anonymisation for data-protection requests

Setup and use

- A single-file Windows application (~57 MB) — no Python installation needed
- Double-click it; on first run it offers to try the system with demo data
- Database, files and logs live in the folder next to the program
- No internet required; cloud AI is optional
- PostgreSQL and Docker support for those who want a server setup

LEARNING CURVE

The application contains a built-in guide of eight lessons. Lessons are filtered by role; a server is not shown the backup lesson. About 38 minutes in total.

Technical foundation

Built to last, to be auditable, and to grow.

- Python 3.11+ and Django 5.2 — mature, documented, widely supported
- WebSockets for the kitchen display (Django Channels, Daphne)
- SQLite on a single machine; PostgreSQL for multi-terminal setups
- 484 automated tests; linter, formatter and security scan on every release
- Dependency vulnerability and secret scanning required before publishing
- All settings come from the environment; no secret lives in the source code

Fiscal compliance notice

The receipts, checks and daily reports this system produces are for internal information and do not replace legal fiscal documents.

- A legal Z report must be produced by a certified fiscal device.
- Electronic invoices must be issued through an authorised integrator.
- The system provides interfaces for these integrations; it does not claim to produce fiscal documents.

Quality measures

Verified automatically before every release.

484

tests

unit, integration, permissions,
security

pip-audit + **1,764**

dependency scan

run before every release

1,764

translated strings

Turkish / English

gitleaks

secret scan

run before every commit

Ready to try

The program runs as a single file and opens with a restaurant already full of sample data.

- 1 Put the application in an empty folder and double-click it
- 2 Choose "Try with sample data"
- 3 The sample-data step prints the usernames and a one-time random password for that install
- 4 Reach the user guide from the ? button in the top bar

Clear the sample data and create your own users before going live.